LinearSkills Academy

Class 4 — Strings

Phase 1: Foundations (Creation & Concatenation)

Goal: Learn how Python recognizes text and how to glue text together.

Key Concept: Strings are wrapped in quotes. You can add them together using +.

```
# 1. Creating Strings
single_quote = 'Hello'
double_quote = "World"
multi_line = """This is a string
that spans multiple lines."""

# 2. String Concatenation (Joining strings)
greeting = single_quote + " " + double_quote
print(greeting)  # Output: Hello World

# 3. String Multiplication (Repeating text)
laugh = "Ha" * 3
print(laugh)  # Output: HaHaHa
```

Phase 2: Indexing and Slicing (Slicing the Cake)

Goal: Extract specific characters or sections of text.

Key Concept: Python starts counting at 0. Negative numbers count backward from the end.

Syntax: string[start : stop : step] (stop is exclusive)

```
topic = "Python"

# 1. Indexing (Getting a single character)
print(topic[0])   # Output: P (First letter)
print(topic[-1])  # Output: n (Last letter)

# 2. Slicing (Getting a chunk)
print(topic[0:4])  # Output: Pyth (Indexes 0, 1, 2, 3)
print(topic[2:])   # Output: thon (From index 2 to the very end)

# 3. Stepping & Reversing
print(topic[::2])  # Output: Pto (Every second letter)
print(topic[::-1]) # Output: nohtyP (Trick to reverse a string!)
```

Phase 3: The Toolkit (Essential String Methods)

Goal: Learn how to modify, clean, and search inside text.

Key Concept: Strings are immutable (they cannot be changed in place). Methods return a new string.

```
text = "  learn python programming  "

# 1. Cleaning up whitespace
clean_text = text.strip()
print(clean_text)  # Output: "learn python programming"

# 2. Changing Case
print(clean_text.upper())      # Output: LEARN PYTHON PROGRAMMING
print(clean_text.title())      # Output: Learn Python Programming

# 3. Search and Replace
print(clean_text.replace("python", "AI"))  # Output: learn AI programming
print("python" in clean_text)              # Output: True (Checks if word exists)

# 4. Splitting and Joining
words = clean_text.split(" ")
print(words)  # Output: ['learn', 'python', 'programming']

joined_text = "-".join(words)
print(joined_text)  # Output: learn-python-programming
```

Phase 4: String Formatting (Dynamic Text)

Goal: Inject variables smoothly into your text.

Key Concept: Use f-strings (formatted string literals). They are the modern Python standard.

```
name = "Ali"
score = 95

# The old way (clunky)
# message = "Hello " + name + ", your score is " + str(score)

# The modern way: f-strings
message = f"Hello {name}, your score is {score}!"
print(message)  # Output: Hello Ali, your score is 95!

# You can even do math inside f-strings
print(f"Next year, you will be {20 + 1} years old.")
```